

```
1 import math
2
3 # f(x)=x**2-2=0
4 # f'(x)=2x
5
6 def Newton(x,eps):
7     if x==0:
8         return False
9     while True:
10         x_old=x
11         x=x-(x**2-2)/(2*x)
12         if abs(x-x_old) < eps:
13             return x
14
15 def bi_search(eps):
16     left,right=1,2
17     while right-left >= eps:
18         mid=(left+right)/2
19         if mid**2 > 2:
20             right=mid
21         else:
22             left=mid
23     return mid
24
25
26
27 if __name__=="__main__":
28     x=Newton(-5,1e-6)
29     y=bi_search(1e-6)
30     print(x,y)
```